

Experiment 7

Slide 6 — Problem Statements

Question 1

Consider a relation R having attributes as $R(A,B,C,D)$, functional dependencies are given below: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. Identify the set of candidate keys possible in relation R . List all the set of prime and non-prime attributes.

Candidate Key(s): AB, BC, BD

Closure Working:

- $AB^+ = A, B, C(AB \rightarrow C), D(C \rightarrow D) = \{A, B, C, D\} \rightarrow \text{full} \Rightarrow AB$ is a key
- $BC^+ = B, C, D(C \rightarrow D), A(D \rightarrow A) = \{A, B, C, D\} \rightarrow \text{full} \Rightarrow BC$ is a key
- $BD^+ = B, D, A(D \rightarrow A), C(AB \rightarrow C, \text{ using } A, B) = \{A, B, C, D\} \rightarrow \text{full} \Rightarrow BD$ is a key

Prime Attribute(s): A, B, C, D (all four attributes appear in at least one candidate key)

Non-Prime Attribute(s): None

Highest Normal Form: **3NF**

Question 2

Relation $R(A,B,C,D,E)$ having functional dependencies as: $A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R . List all the set of prime and non-prime attributes.

Candidate Key(s): AC, BC

Closure Working:

- $AC^+ = A, C, B, E(AC \rightarrow BE), D(A \rightarrow D) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow AC$ is a key
- $BC^+ = B, C, D(BC \rightarrow D), A(B \rightarrow A), E(AC \rightarrow BE, \text{ using } A, C) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow BC$ is a key

Prime Attribute(s): A, B, C

Non-Prime Attribute(s): D, E

Highest Normal Form: **1NF**

Question 3

Consider a relation R having attributes as $R(A,B,C,D,E)$, functional dependencies are given below: $B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R . List all the set of prime and non-prime attributes.

Candidate Key(s): B, AC (two independent candidate keys)

Closure Working:

- $B^+ = B, A(B \rightarrow A), C(A \rightarrow C), D(BC \rightarrow D, \text{ using } B, C), E(AC \rightarrow BE, \text{ using } A, C) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow B \text{ alone is a key}$
- $AC^+ = A, C, B(AC \rightarrow BE), E(AC \rightarrow BE), D(BC \rightarrow D, \text{ using } B, C) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow AC \text{ is a key (and does not contain } B)$

Prime Attribute(s): A, B, C

Non-Prime Attribute(s): D, E

Highest Normal Form: **3NF**

Question 4

Consider a relation R having attributes as $R(A, B, C, D, E, F)$, functional dependencies are given below: $A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$. Identify the set of candidate keys possible in relation R . List all the set of prime and non-prime attributes.

Candidate Key(s): AF, BF, DF

Closure Working:

- $AF^+ = A, F, B, C, D(A \rightarrow BCD), E(BC \rightarrow DE) = \{A, B, C, D, E, F\} \rightarrow \text{full} \Rightarrow AF \text{ is a key}$
- $BF^+ = B, F, D(B \rightarrow D), A(D \rightarrow A), C(A \rightarrow BCD), E(BC \rightarrow DE) = \{A, B, C, D, E, F\} \rightarrow \text{full} \Rightarrow BF \text{ is a key}$
- $DF^+ = D, F, A(D \rightarrow A), B, C(A \rightarrow BCD), E(BC \rightarrow DE) = \{A, B, C, D, E, F\} \rightarrow \text{full} \Rightarrow DF \text{ is a key}$

Prime Attribute(s): A, B, D, F

Non-Prime Attribute(s): C, E

Highest Normal Form: **1NF**

Slide 24 — Practice Set

Question 1

Relation $R(A,B,C,D,E)$ having dependencies as: $CE \rightarrow D$, $D \rightarrow B$, $C \rightarrow A$. Highest normal form? Candidate Key set?

Candidate Key(s): CE (C and E never appear on the RHS of any FD, so both are essential)

Closure Working:

- $CE^+ = C, E, D(CE \rightarrow D), B(D \rightarrow B), A(C \rightarrow A) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow CE$ is the (only) key

Prime Attribute(s): C, E

Non-Prime Attribute(s): A, B, D

Highest Normal Form: **1NF only**

Reasoning:

- $C \rightarrow A$: C is a proper subset of candidate key CE, C is not a superkey ($C^+ = \{A, C\}$), and A is non-prime $\Rightarrow$ partial dependency.
- This violates 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.

Question 2

Relation $R(A,B,C,D,E,F)$ having dependencies as: $AB \rightarrow C$, $DC \rightarrow AE$, $E \rightarrow F$. Highest normal form? Candidate Key set?

Candidate Key(s): ABD, BCD (B and D never appear on the RHS of any FD, so both are essential; a third attribute — A or C — is needed to complete the closure)

Closure Working:

- $ABD^+ = A, B, D, C(AB \rightarrow C), E(DC \rightarrow AE), F(E \rightarrow F) = \{A, B, C, D, E, F\} \rightarrow \text{full} \Rightarrow ABD$ is a key
- $BCD^+ = B, C, D, A, E(DC \rightarrow AE), F(E \rightarrow F) = \{A, B, C, D, E, F\} \rightarrow \text{full} \Rightarrow BCD$ is a key

Prime Attribute(s): A, B, C, D

Non-Prime Attribute(s): E, F

Highest Normal Form: **1NF only**

Reasoning:

- $DC \rightarrow AE$: CD is a proper subset of candidate key BCD, CD is not a superkey ($CD^+ = \{A, C, D, E, F\}$, missing B), and E is non-prime $\Rightarrow$ partial dependency.
- This violates 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.

Question 3

Relation $R(A,B,C,D,E,F,G,H,I)$ having dependencies as: $AB \rightarrow C$, $BD \rightarrow EF$, $AD \rightarrow GH$, $A \rightarrow I$. Highest normal form? Candidate Key set?

Candidate Key(s): ABD (A, B, D never appear on the RHS of any FD, so all three are essential and together they are sufficient)

Closure Working:

- $ABD^+ = A, B, D, C(AB \rightarrow C), E, F(BD \rightarrow EF), G, H(AD \rightarrow GH), I(A \rightarrow I) = \{A, B, C, D, E, F, G, H, I\} \rightarrow \text{full} \Rightarrow ABD$ is the (only) key

Prime Attribute(s): A, B, D

Non-Prime Attribute(s): C, E, F, G, H, I

Highest Normal Form: **1NF only**

Reasoning:

- $AB \rightarrow C$: AB is a proper subset of candidate key ABD, AB is not a superkey ($AB^+ = \{A, B, C, I\}$, missing D, E, F, G, H), and C is non-prime $\Rightarrow$ partial dependency.
- This violates 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.

Question 4

Relation $R(A, B, C, D, E)$ having dependencies as: $AB \rightarrow CD$, $D \rightarrow A$, $BC \rightarrow DE$. Highest normal form? Candidate Key set?

Candidate Key(s): AB, BC, BD (B never appears on the RHS of any FD, so B is essential in every key)

Closure Working:

- $AB^+ = A, B, C, D(AB \rightarrow CD), E(BC \rightarrow DE) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow AB$ is a key
- $BC^+ = B, C, D, E(BC \rightarrow DE), A(D \rightarrow A) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow BC$ is a key
- $BD^+ = B, D, A(D \rightarrow A), C(AB \rightarrow CD, \text{ using } A, B) = B, D, A, C$ then $BC \rightarrow DE$ gives $E = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow BD$ is a key

Prime Attribute(s): A, B, C, D

Non-Prime Attribute(s): E

Highest Normal Form: **3NF (not BCNF)**

Reasoning:

- $AB \rightarrow CD$: AB is a candidate key (superkey) $\Rightarrow$ OK for BCNF.
- $D \rightarrow A$: D is not a superkey ($D^+ = \{A, D\}$), but A is a prime attribute $\Rightarrow$ satisfies 3NF, violates BCNF.
- $BC \rightarrow DE$: BC is a candidate key (superkey) $\Rightarrow$ OK for BCNF.
- No non-prime attribute depends on a proper subset of a key (2NF holds); only $D \rightarrow A$ breaks BCNF, so the relation is in 3NF.

Question 5

Relation $R(A, B, C, D, E)$ having dependencies as: $BC \rightarrow ADE$, $D \rightarrow B$. Highest normal form? Candidate Key set?

Candidate Key(s): BC, CD (C never appears on the RHS of any FD, so C is essential in every key)

Closure Working:

- $BC^+ = B, C, A, D, E (BC \rightarrow ADE) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow BC$ is a key
- $CD^+ = C, D, B (D \rightarrow B), A, E (BC \rightarrow ADE, \text{ using } B, C) = \{A, B, C, D, E\} \rightarrow \text{full} \Rightarrow CD$ is a key

Prime Attribute(s): B, C, D

Non-Prime Attribute(s): A, E

Highest Normal Form: **3NF (not BCNF)**

Reasoning:

- $BC \rightarrow ADE$: BC is a candidate key (superkey) $\Rightarrow$ OK for BCNF.
- $D \rightarrow B$: D is not a superkey ($D^+ = \{B, D\}$), but B is a prime attribute $\Rightarrow$ satisfies 3NF, violates BCNF.
- No non-prime attribute depends on a proper subset of a key (2NF holds); only $D \rightarrow B$ breaks BCNF, so the relation is in 3NF.

Question 6

Relation $R(A, B, C)$ having dependencies as: $A \rightarrow B, B \rightarrow C, C \rightarrow A$. Highest normal form? Candidate Key set?

Candidate Key(s): A, B, C (each single attribute is itself a candidate key — a circular FD chain)

Closure Working:

- $A^+ = A, B (A \rightarrow B), C (B \rightarrow C) = \{A, B, C\} \rightarrow \text{full} \Rightarrow A$ is a key
- $B^+ = B, C (B \rightarrow C), A (C \rightarrow A) = \{A, B, C\} \rightarrow \text{full} \Rightarrow B$ is a key
- $C^+ = C, A (C \rightarrow A), B (A \rightarrow B) = \{A, B, C\} \rightarrow \text{full} \Rightarrow C$ is a key

Prime Attribute(s): A, B, C (all attributes are prime)

Non-Prime Attribute(s): None

Highest Normal Form: **BCNF**

Reasoning:

- $A \rightarrow B$: determinant A is a candidate key (superkey) $\Rightarrow$ OK.
- $B \rightarrow C$: determinant B is a candidate key (superkey) $\Rightarrow$ OK.
- $C \rightarrow A$: determinant C is a candidate key (superkey) $\Rightarrow$ OK.
- Every FD's determinant is a superkey, so the relation satisfies BCNF (the highest normal form).

Slide 25 — Problem Statement

Question

Consider a relation P having fields as $F1, F2, F3, F4, F5$ with the functional dependencies as follows: $F1 \rightarrow F3, F2 \rightarrow F4, (F1, F2) \rightarrow F5$. In terms of normalization, this table is in which normal form? Describe the prime and non-prime attributes for the relation P .

Candidate Key(s): $F1F2$ ($F1$ and $F2$ never appear on the RHS of any FD, so both are essential and together are sufficient)

Closure Working:

- $(F1, F2)^+ = F1, F2, F3(F1 \rightarrow F3), F4(F2 \rightarrow F4), F5((F1, F2) \rightarrow F5) = \{F1, F2, F3, F4, F5\} \rightarrow \text{full} \Rightarrow F1F2$ is the (only) key

Prime Attribute(s): $F1, F2$

Non-Prime Attribute(s): $F3, F4, F5$

Highest Normal Form: **1NF only**

Reasoning:

- $F1 \rightarrow F3$: $F1$ is a proper subset of candidate key $F1F2$, $F1$ is not a superkey, and $F3$ is non-prime $\Rightarrow$ partial dependency.
- $F2 \rightarrow F4$: similarly, $F2$ is a proper subset of the key, not a superkey, and $F4$ is non-prime $\Rightarrow$ another partial dependency.
- These partial dependencies violate 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.

Slide 26 — Practice Set

Question 1

Consider a relation $R(A,B,C,D,E,F,G,H,I)$ having FDs as: $AB \rightarrow C$, $AD \rightarrow GH$, $BD \rightarrow EF$, $A \rightarrow I$, $H \rightarrow J$. Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.

Note: R is listed with attributes $A-I$, but the FD " $H \rightarrow J$ " refers to an attribute J that is not part of R . This is most likely a typo in the original slide. The solution below treats R as $R(A,B,C,D,E,F,G,H,I)$ and works with the FDs that involve only R 's own attributes ($AB \rightarrow C$, $AD \rightarrow GH$, $BD \rightarrow EF$, $A \rightarrow I$); " $H \rightarrow J$ " does not affect the key computation for R .

Candidate Key(s): ABD (A, B, D never appear on the RHS of any FD, so all three are essential and together they are sufficient)

Closure Working:

- $ABD^+ = A,B,D,C(AB \rightarrow C),G,H(AD \rightarrow GH),E,F(BD \rightarrow EF),I(A \rightarrow I) = \{A,B,C,D,E,F,G,H,I\} \rightarrow \text{full} \Rightarrow ABD$ is the (only) key

Prime Attribute(s): A, B, D

Non-Prime Attribute(s): C, E, F, G, H, I

Highest Normal Form: **1NF only**

Reasoning:

- $AB \rightarrow C$: AB is a proper subset of candidate key ABD, AB is not a superkey, and C is non-prime $\Rightarrow$ partial dependency.
- This violates 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.

Question 2

Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows: $A \rightarrow B$, $B \rightarrow E$, $C \rightarrow D$. Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.

Candidate Key(s): AC (A and C never appear on the RHS of any FD, so both are essential)

Closure Working:

- $AC^+ = A,C,B(A \rightarrow B),D(C \rightarrow D),E(B \rightarrow E) = \{A,B,C,D,E\} \rightarrow \text{full} \Rightarrow AC$ is the (only) key

Prime Attribute(s): A, C

Non-Prime Attribute(s): B, D, E

Highest Normal Form: **1NF only**

Reasoning:

- $A \rightarrow B$: A is a proper subset of candidate key AC, A is not a superkey, and B is non-prime $\Rightarrow$ partial dependency.
- $C \rightarrow D$: similarly, C is a proper subset of the key, not a superkey, and D is non-prime $\Rightarrow$ another partial dependency.
- These partial dependencies violate 2NF, so the relation cannot reach 3NF or BCNF $\Rightarrow$ only 1NF holds.